



Trash Talk: Alleviating the Growing Menace of Space Junk

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The presence of human-made debris orbiting the planet Earth is now a problem for space agencies and research organisations like NASA, ISRO and ESA. With the volume of floating 'space junk' nearing the 7000-tonne mark, concerns are starting to mount and effective methods for removing it now need to be considered.

Space junk basically comprises a whole host of defunct objects - the remains of old satellites, spent rocket stages and the countless fragments left over from disintegration, erosion, collisions, etc. This flotsam and jetsam poses a serious threat to satellites and spacecraft. Travelling at speeds as high as 35,000mph, even the smallest fragments have the potential to cause serious damage. Scientists universally agree that the issue can no longer be ignored and if a solution is not found soon it will impinge upon the continued advancement of space exploration.

With nearly 700,000 objects measuring greater than 1cm in size currently in orbit, combined with the high velocities at which they are moving, the risk of their colliding into something is a significant one. NASA, together with the US Department of Defence, are attempting to monitor the position and trajectory of items of space junk - with over 20,000 objects currently being tracked. However it is difficult to do this for anything but the larger pieces. Recent instances of satellites being struck by objects too small to be tracked (and damage resulting) throw a spotlight on the intricacies of the problem. Add to this the advent of space tourism and the expectation of more people making flights beyond the Earth's atmosphere and safety has to start being questioned.

The amount of space debris orbiting Earth will keep increasing. This, in turn, is raising the probability of more collisions - thereby generating yet more orbiting junk. Then, as if it couldn't get any busier up there, a substantial drop in the manufacturing costs of building satellites has heralded the beginning of the production of satellite mega-constellations. This is set to lead to the deployment of hundreds, if not thousands, of new satellites into low-Earth orbits in the near future - thereby adding more potential targets for space junk to hit. Space debris orbiting the earth poses a threat to safe space travel.

The RemoveDebris project, however, may be about to tip the balance back in our favour. The project aims to tackle the growing threat of space junk by testing a range of possible junk removal devices in space. With financial backing from the European Com-

mission, an international consortium led by the Surrey Space Centre (SSC) and including aerospace manufacturer Airbus, together with leading research and industrial institutions from France, Switzerland and South Africa, is undertaking an extensive study of a variety of methods and novel technologies to catch and dispose of pieces of floating space debris. Among the proposed methods is the deceptively simple idea of using a large net to haul in space junk. This would be deployed by a satellite ejected from the 'mothership'. Alternatively harpoons could be a way to capture large items. The harpooned or netted objects would then be dragged along by the mothership and eventually back down towards Earth and deorbited, at which point they would burn up in the atmosphere.

During upcoming tests, the intention is for nets and harpoons to be fired at some of the SSC's CubeSats - miniature satellites that will act as artificial space junk for the purposes of the investigation. The findings of the RemoveDebris project will be of significant value to the long-term progress of space exploration, paving the way for safer spaceflight and ensuring that satellites and orbiting space stations have some defence against the threat of incapacitating damage and the risk to life that space junk presents. The extremes of space and the harsh operational conditions of working beyond the Earth's orbit, such as the impact of shock and vibration experienced during launch together with exposure to extremes of temperature, require that all satellite-based technology needs to be exceptionally robust. Add to this the complexities of employing SSC's ultra-compact CubeSats and the engineering challenges of identifying the most suitable constituent components for such a project are immense.

For some six years now, high-reliability connector supplier Harwin has worked with SSC on the development of its state-of-the-art CubeSat technology, supplying it with connectivity components from the highly advanced Gecko range. Exhibiting the ruggedness required to maintain continuing operational reliability, these connectors are also small and lightweight enough for SSC's precise requirements. Launching to the International Space Station on board a SpaceX rocket, in-space testing of the RemoveDebris project is set to take place early next year. In doing so this will become the first mission of its kind to test debris removal technology in space; literally clearing the path for the future of space endeavour.

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